ANSWER KEY

Ling 285

Exam 1, 10/1/2026

Please fill in your name legibly at the top. There are 25 questions. They all have the same number of points, and you need to answer all of them. Enter all your answers on the exam sheet. ***You cannot use your phone, computer watch, or computer during the exam***. The space after each question is suggestive of the number of sentences you should write. You can leave after you're done with the exam.

1. Which of these 4 could be the probability distribution decision of a neural network? _ **b** _

   a. [0,0,0,0] **b. [0,.4,0,.6]** c. [-.5,0,.5,0] d. [.3,.3,.4,.5]

   Only (b) is possible. The numbers in a probability distribution have to be between 0 and 1 and all add up to 1. (a) adds up to 0, (c) has a negative number, and (d) adds up to 1.5.

2. Name three ways in which we can distinguish consonants from each other.

   Voicing (whether the vocal folds vibrate), place of articulation (where in the mouth the sound is made, e.g., labial, dental, alveolar, postalveolar, or velar), manner of articulation (how the air is constricted, e.g., fricative, oral stop, or nasal stop), and nasality (whether the velum is lowered to allow the air to escape through the nose). Any 3 of the four.

3. One side of the basilar membrane is heavy and not stiff, while the other side is light and very stiff. How do these facts enable audition?

   Frequency goes up with stiffness and down with mass (k↑ f↑, m↑ f↓), so the heavy, non-stiff end of the membrane vibrates at low frequencies and the light, stiff end vibrates at high frequencies. Each part is hooked to neurons that tell the brain which frequencies they heard based on location.

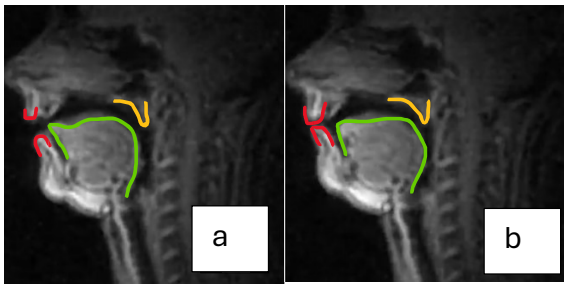
4. Can you describe how dynamic programming works in terms of the size of problems, and problems containing other problems

   You break the big problem into smaller subproblems until you reach small problems that are trivial to solve. You solve those smallest problems first, then use their answers to solve the larger problems that contain them, all the way to the hardest problems.

5. Can you describe why dynamic programming is more efficient than the exhaustive algorithm**?**

   The paths in a shortest-path problem overlap. The exhaustive algorithm solves every path separately, so it redoes the same additions over and over. Dynamic programming solves only the non-overlapping parts. It finds the best solution from each state only once and reuses it for every path through that state, so it never repeats work.

6. Which of these is “f” as in “family” __ **b** __ and which is “th” as in “thing” __ **a** ___



   In (b) the lower lip is against the upper teeth (labial); in (a) the tongue tip is up at the teeth (dental).

7. Describe how the Stochastic Gradient Descent Algorithm works in your own words.

   We want to estimate a quantity from some existing data, like the average of a list of numbers. We start with a random estimate, pick a random data point, and compute the loss (the difference between our guess and that data point). Then we subtract a small fraction of the loss (the learning rate) from the guess to get a new, slightly better guess. We repeat this with new random data points over and over, and the guess gets closer and closer to the real answer.

8. What's the similarity between "n" in "neck" and "d" in "deck", and what's the difference between those sounds?
   Both are voiced alveolar stops. The difference is that n is a nasal stop, so the air comes out through the nose, while d is an oral stop, so the air is blocked completely and then released through the mouth.

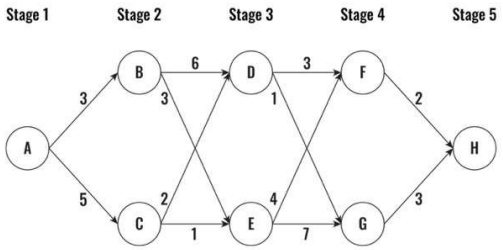
9. What's the similarity and difference between the vowels in "mean" and "moon"?
   Both are high vowels. The difference is that the vowel in "mean" is a front vowel, while the vowel in "moon" is a back vowel (lips also rounded).

10. Imagine a Stochastic Gradient Algorithm (SGD) with a very low learning rate and a springy system (with no mass) with a low stiffness. a) Describe how the Estimate of SGD and the equilibrium position for the spring system evolve with time. b) How would the evolution in time differ, if the learning rate and stiffness increased? For SGD, ignore accuracy/oscillation for this question.
    a) Both move toward their goal slowly. With a low learning rate, the SGD estimate only takes a small correction toward the average each time, so it takes many steps to get there. With low stiffness, the spring only pulls weakly, so it drifts back to its equilibrium position slowly.

    b) Both would get there faster. A higher learning rate makes bigger corrections toward the average, and a stiffer spring pulls harder and snaps back to equilibrium more quickly.

11. Describe the psychological principle of closure, illustrated by this figure  , and how is it used in the GPT system as a principle for learning?
    Humans are very adept at filling in missing information using what we already know. In the figure no triangle is actually drawn, but we see one anyway because our brain completes the shape. GPT learns the same way (the cloze task). It takes a sentence, pretends a part of a sentence is missing, and asks the network to complete what's missing. If the guess is wrong, the weights get adjusted, and after billions of these completions the system ends up "knowing" the language.

12. Referring to the shortest path problem below, what is the Greedy Forward solution? You can just write out the numbers with +'s. ____**3+3+4+2 = 12 (A→B→E→F→H)**____



13. Referring to the shortest path problem in question 12. We call the best cost from any state x to state H V_{xH}. Imagine we have already solved for V_{DH} and V_{EH}, how can I calculate V_{CH}? **V_{CH} = min{ 2 + V_{DH}, 1 + V_{EH} } (or written vertically, crucially *min*)**

14. In terms of accuracy and speed in solving shortest path problems, how do Greedy and Exhaustive Algorithms compare?
    Greedy is fast but inaccurate. Exhaustive is accurate but slow.

    And what algorithm do we use if we want to balance those two factors?
    ___**Dynamic Programming**___

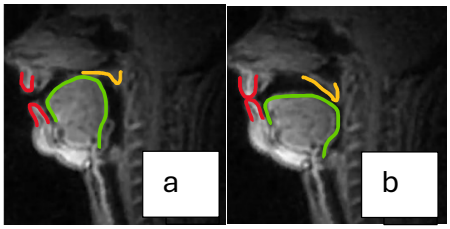
15. What's my favorite popular way of describing how learning in AI works? **Fake it till you make it**. Describe why I put it this way.
    The system starts out with completely random weights, so it doesn't actually know anything. It's just faking it. It makes a guess, receives feedback, and changes its weights a little so it does better next time. After repeating this a lot of times, it will make it, i.e., have the right knowledge.

16. In each vowel, air vibrates at a single frequency? Yes/**No** ___
17. What is the goal of a spring, if it's extended or squeezed? ___**Go back to equilibrium**_ .
    What is the primary physical property of a mass that is already moving?
    ___**keeps on moving**___

18. If a system is described by these differential equations $\begin{matrix}\dot{x} = v \\ \dot{v} = -\frac{k}{m}x\end{matrix}$, and if x starts as negative and v = 0, will the system oscillate? How will the position of the mass evolve with time.
    Yes, it oscillates. x starts out negative, so the spring is squeezed and pushes the mass toward equilibrium (x = 0). The mass speeds up and reaches 0, but a moving mass keeps moving, so it overshoots and stretches the spring on the positive side;

the spring then pulls it back, it overshoots again, and this repeats. The position swings back and forth, tracing out a cosine-shaped wave.

19. Which of these is "k" as in "Ken" ___**a**____ and which of these is "m" as in "men"?_**b**__



In (a), the back of the tongue is raised up (velar); in (b), the lips are closed (labial) and the velum is also lowered (nasal).

20. In SGD for averaging a list, the mathematical formula for the New Estimate of average x is: New x = Current x – lr (Current x – data point d to learn from). And lr is the learning rate. What is the purpose of the first minus in the formula?

    The minus makes the estimate move toward the data point regardless if the data is above or below the true estimate. If the estimate is above the data point, the loss is positive, so subtracting it pushes the estimate down. If the estimate is below the data point, the loss is negative, so subtracting it pushes the estimate up.

21. Imagine we have a neural network with many neurons in many layers. Imagine that one neuron, let's call it y, in layer 4 is connected to 5 neurons from the previous layer. Assume every neuron is connected to all neurons in previous layer. How many weights does y have? _**5**__
22. Continuing from the previous question, if y's weights are w1, w2, w3, w4, w5, and the activations of the 5 neurons feeding it are x1, x2, x3, x4, x5. How do we write the activation of neuron y? __**w1x1+w2x2+w3x3+w4x4+w5x5**_____

23. Imagine that the decision made by a neural network is [.4,.1,.1,.3,.1], but the true category is the one in the last position. Using Python list notation, how do we express to the neural network in a probability distribution format what the true answer is? _____**[0,0,0,0,1]**_____

24. If I raise stiffness of a mass-spring system, does the frequency of vibration rise or fall? _**rise**_ And if I raise the mass, does the frequency of vibration rise or fall? _**fall**___

25. There are systems all around us? True or False? __**True**__